

Adaptive Sub-Prototype Learning for Long-Tailed Visual Recognition

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Abstract—Prototypical classification methods represent each category by a single feature center, achieving strong performance on balanced data. Under long-tailed distributions, however, this single-prototype paradigm encounters three coupled deficiencies: (i) at the *feature level*, head classes exhibit multi-modal intra-class structure that a single point cannot capture; (ii) at the *update level*, online prototype estimates for tail classes suffer from high variance due to sample scarcity; (iii) at the *decision level*, the categorical frequency prior systematically shifts the decision boundary against tail classes. We propose Adaptive Sub-Prototype Network (ASPNet), which addresses these deficiencies through three tightly coupled components: (1) an adaptive sub-prototype allocation strategy that determines the number of sub-prototypes per class via a logarithmic function of class cardinality; (2) a divergence-aware confidence gating mechanism that quantifies mini-batch estimation reliability and employs Hungarian matching to maintain temporal consistency; (3) a prior-calibrated prototype classifier that aggregates sub-prototype similarities through LogSumExp pooling and corrects logits with the categorical frequency prior. Experiments on CIFAR-10-LT, CIFAR-100-LT, ImageNet-LT, and Places-LT demonstrate that ASPNet achieves competitive or superior performance with a minimalist two-term loss function.

Index Terms—Long-tailed recognition, prototype learning, sub-prototype, confidence gating, logit calibration

I. INTRODUCTION

Deep visual recognition has achieved remarkable progress under the assumption of approximately balanced training distributions [16], [17]. Real-world data, however, inherently follows a long-tailed distribution: a small number of head classes contain abundant samples, while a large number of tail classes are severely under-represented. Standard empirical risk minimization on such data yields classifiers that are systematically biased toward head classes, resulting in substantial accuracy degradation on tail classes [7], [8].

A rich body of work has been developed to mitigate this bias, spanning re-sampling, cost-sensitive learning, logit adjustment, decoupled training, and contrastive learning [18], [19]. Among these, prototype-based methods are particularly appealing: they replace the fully connected classifier with distance-based classification against learned class prototypes, naturally integrating with metric learning and offering a parameter-efficient alternative [3], [5]. Nevertheless, the prevailing “one prototype per class” paradigm harbors a fundamental limitation in long-tailed settings [2].

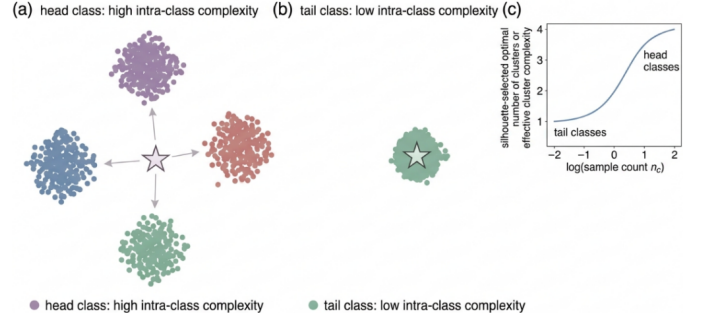


Fig. 1. Imbalanced intra-class complexity under long-tailed distributions. (a) Head classes exhibit multi-modal feature distributions with high intra-class diversity. (b) Tail classes concentrate around a single mode with low intra-class diversity. (c) Silhouette-selected optimal cluster count increases monotonically with $\log(n_c)$, confirming that intra-class structural complexity scales with sample count. Results obtained from ResNet-32 features on CIFAR-10-LT (IF = 100).

A. Core Observation

To expose this limitation, we train a standard ResNet-32 on CIFAR-10-LT (imbalance factor IF = 100), extract training-set features, and evaluate the optimal cluster count per class using the Silhouette Score. As illustrated in Fig. 1, a pronounced pattern emerges: head classes ($n_c = 5000$) naturally partition into 3–4 visually coherent sub-groups, whereas tail classes ($n_c = 50$) concentrate into 1–2 sub-groups. This reveals an *imbalanced intra-class structural complexity*—a dimension that existing long-tailed methods rarely model explicitly [19].

This observation gives rise to three cascading challenges. **(C1) Feature-level: intra-class structure mismatch.** Collapsing a multi-modal head-class distribution into a single prototype discards sub-group structure and compresses the discriminative margin available to neighboring tail classes (Fig. 2). **(C2) Update-level: unstable tail prototypes.** When only a handful of tail-class samples appear in a mini-batch, the online prototype estimate exhibits high variance, causing semantic drift across iterations. **(C3) Decision-level: biased decision boundary.** The imbalanced categorical prior shifts the decision boundary toward tail classes, further suppressing their recall [6].

To jointly address C1–C3, we propose **Adaptive Sub-Prototype Network (ASPNet)**. The guiding principle is threefold: *the number of sub-prototypes should match intra-class complexity, the update should match estimation re-*

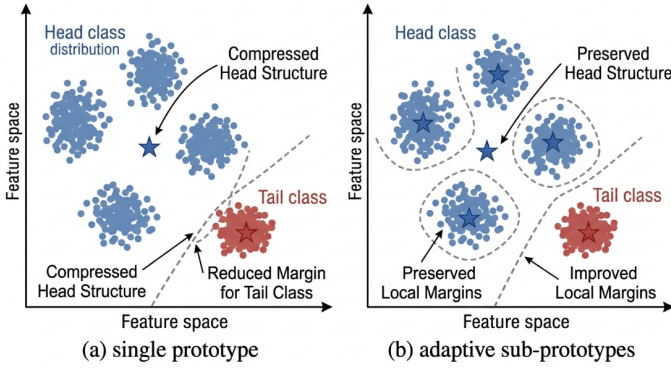


Fig. 2. Geometric comparison of single-prototype and adaptive sub-prototype representations in feature space. (a) A single prototype compresses the multi-modal head-class distribution, reducing the decision margin for the adjacent tail class. (b) Adaptive sub-prototypes preserve head-class structure and enlarge local margins for both head and tail classes.

liability, and the classification should match the balanced posterior. Concretely, ASPNet comprises: (1) an adaptive sub-prototype allocation that assigns more sub-prototypes to complex head classes and fewer to tail classes (addressing C1); (2) a divergence-aware confidence gating update with Hungarian matching that stabilizes tail-class prototypes while allowing head-class prototypes to adapt rapidly (addressing C2); (3) a prior-calibrated prototype classifier that aggregates sub-prototype similarities via LogSumExp and corrects the categorical prior bias (addressing C3).

Our contributions are as follows:

- We identify and empirically characterize the imbalanced intra-class structural complexity in long-tailed distributions, and propose an adaptive sub-prototype allocation strategy governed by a logarithmic function of class cardinality.
- We design a divergence-aware confidence gating mechanism that quantifies mini-batch estimation reliability and combines it with Hungarian matching to ensure temporally consistent sub-prototype updates.
- We demonstrate that ASPNet, equipped with only a two-term loss function, achieves competitive or state-of-the-art performance on four standard long-tailed benchmarks.

II. RELATED WORK

A. Long-Tailed Visual Recognition

Existing long-tailed recognition methods can be organized along four axes. **Re-sampling and re-weighting** methods adjust the training distribution or per-sample loss weights to counteract class imbalance; representative approaches include Focal Loss [1] and LDAM-DRW [8]. **Logit adjustment** methods compensate the categorical prior directly in the logit space, as exemplified by Balanced Softmax [7] and Logit Adjustment [6]. **Decoupled training** methods separate representation learning from classifier learning, enabling each stage to adopt distribution-appropriate strategies; cRT [4] and MiSLAS [9] are notable instances. **Contrastive learning** methods construct

positive and negative pairs to learn discriminative features robust to class imbalance, including PaCo [11] and BCL [12]. A common assumption across all four families is that each class is adequately represented by a single feature center, which neglects the heterogeneous intra-class complexity that arises under long-tailed distributions.

B. Prototype Networks and Multi-Prototype Learning

Prototypical Networks [3] introduced the idea of computing class-mean features as prototypes for metric-based classification, originally in the few-shot learning context. LPC-LTR [5] extended prototype classifiers to long-tailed recognition. More recently, Guo *et al.* [2] explored fixed multi-prototype representations, but their allocation is class-agnostic and does not adapt to the varying intra-class complexity induced by long-tailed distributions. Our work fills this gap by dynamically allocating sub-prototypes proportional to intra-class complexity and coupling the allocation with a reliability-aware update mechanism.

C. Positioning of This Work

Unlike prior methods that treat intra-class structure as homogeneous across classes, ASPNet explicitly models the *imbalance* in intra-class structural complexity. It unifies sub-prototype allocation, confidence-gated updating, and prior-calibrated classification into a single coherent framework—to our knowledge, the first such unified “allocation–update–classification” prototype framework designed for long-tailed recognition.

III. METHODOLOGY

A. Problem Formulation and Overview

Notation. Let $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N$ denote the training set, where $y_i \in \{1, \dots, C\}$. The sample count of class c is n_c , with $n_1 \geq n_2 \geq \dots \geq n_C$ and imbalance factor $\text{IF} = n_1/n_C \gg 1$.

Feature extraction. A backbone network $f(\cdot)$ extracts features, which are then projected and L_2 -normalized onto the unit hypersphere $\mathbb{S}^{d-1}$:

$$z_i = \text{Norm}(Wf(x_i) + b), \quad \|z_i\|_2 = 1, \quad (1)$$

where W and b are learnable projection parameters.

Fig. 3 provides an overview of the ASPNet pipeline, which consists of three modules detailed below.

B. Adaptive Sub-Prototype Allocation and Initialization

A single prototype cannot capture the multi-modal structure of head classes (challenge C1), yet assigning excessive sub-prototypes to tail classes risks overfitting to scarce samples. We therefore allocate sub-prototypes dynamically based on intra-class complexity, using sample count as a proxy.

Sub-prototype count allocation. Following the principle of logarithmic marginal diminishing returns, the number of sub-prototypes for class c is:

$$K_c = \min\left(K_{\max}, \max(1, \lfloor \alpha \log(n_c + 1) + \beta \rfloor)\right), \quad (2)$$

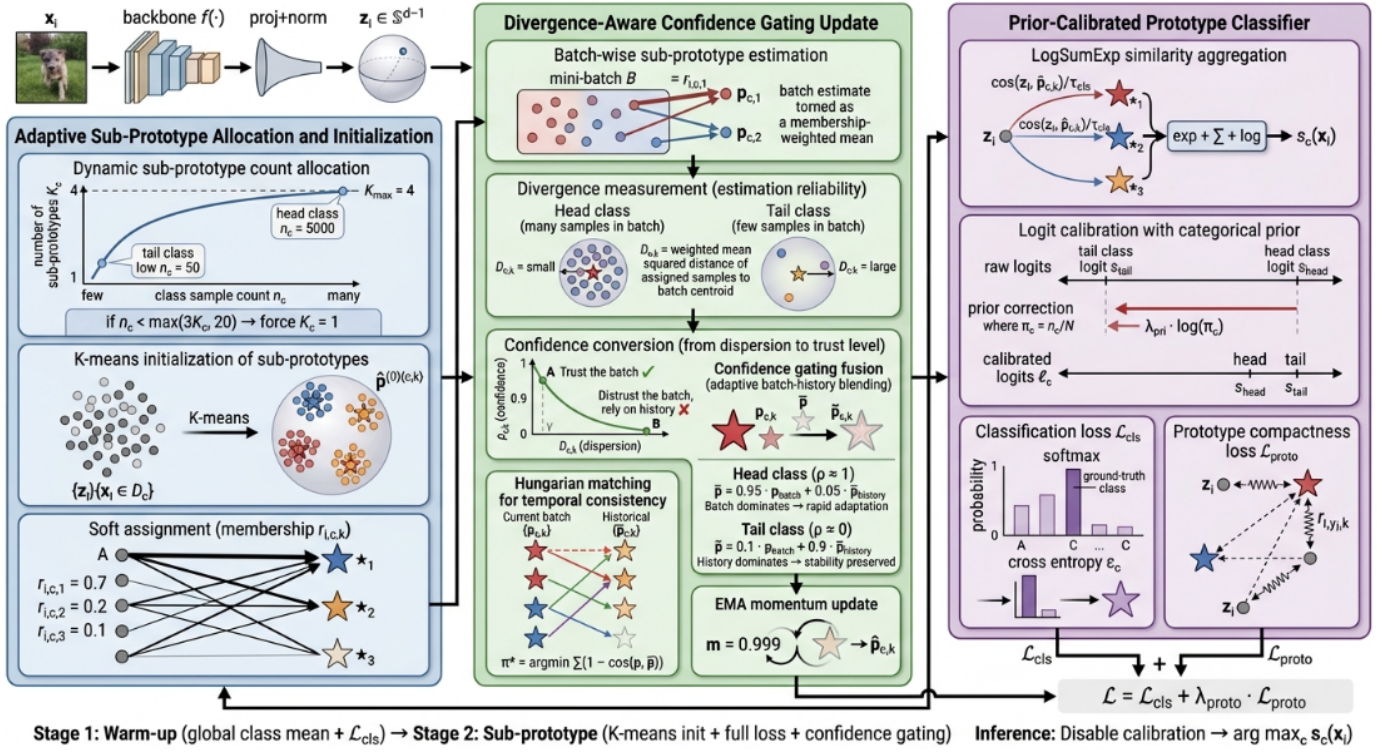


Fig. 3. Overview of the ASPNet framework. **Left:** Adaptive sub-prototype allocation determines K_c per class via a logarithmic function of n_c , followed by K-means initialization and soft assignment. **Center:** Divergence-aware confidence gating update computes batch-wise sub-prototype estimates, measures estimation reliability through weighted dispersion, converts dispersion to confidence scores, applies Hungarian matching for temporal consistency, and performs confidence-gated EMA fusion. **Right:** Prior-calibrated prototype classifier aggregates sub-prototype similarities via LogSumExp, applies logit calibration with the categorical prior, and optimizes the combined classification and prototype compactness losses. Training proceeds in two stages: warm-up with global class means, followed by sub-prototype optimization with the full loss. At inference, logit calibration is disabled and prediction is $\arg \max_c s_c(x_i)$.

where α and β control the curve shape, and $K_{\max}$ is an upper bound. A safety constraint is imposed: if $n_c < \max(3K_c, 20)$, we force $K_c = 1$ to ensure that each sub-prototype is supported by a sufficient number of samples for reliable estimation.

K-means initialization. After a warm-up training phase, we freeze the backbone and initialize sub-prototypes via K-means clustering on the class-conditional feature sets:

$$\hat{p}_{c,k}^{(0)} = \text{K-means}_k(\{z_i\}_{x_i \in \mathcal{D}_c}), \quad (3)$$

where all prototypes are subsequently L_2 -normalized.

Soft assignment. The membership weight of sample i to sub-prototype k of class c is computed via a temperature-scaled softmax over cosine similarities:

$$r_{i,c,k} = \frac{\exp(\langle z_i, \hat{p}_{c,k} \rangle / \tau_q)}{\sum_{k'=1}^{K_c} \exp(\langle z_i, \hat{p}_{c,k'} \rangle / \tau_q)}. \quad (4)$$

C. Divergence-Aware Confidence Gating Update

Online sub-prototype updates face a fundamental tension: head-class estimates from large mini-batch subsets are reliable and should drive rapid adaptation, whereas tail-class estimates from sparse samples are noisy and should be damped (challenge C2). We resolve this tension by quantifying estimation

reliability and gating the update accordingly. Fig. 4 illustrates the four components of this mechanism.

Batch-wise sub-prototype estimation. For class c in mini-batch $\mathcal{B}$, the batch estimate of sub-prototype k is the membership-weighted mean:

$$p_{c,k} = \frac{\sum_{x_i \in \mathcal{B}_c} r_{i,c,k} z_i}{\sum_{x_i \in \mathcal{B}_c} r_{i,c,k} + \varepsilon}. \quad (5)$$

Divergence-driven confidence quantification. We measure estimation reliability via the weighted dispersion of assigned samples around the batch estimate:

$$D_{c,k} = \frac{\sum_{x_i \in \mathcal{B}_c} r_{i,c,k} \|z_i - p_{c,k}\|_2^2}{\sum_{x_i \in \mathcal{B}_c} r_{i,c,k} + \varepsilon}. \quad (6)$$

A small $D_{c,k}$ indicates that the assigned samples are tightly clustered around the estimate, implying high reliability. The dispersion is converted to a confidence score $\rho_{c,k} \in [0, 1]$:

$$\rho_{c,k} = \exp(-D_{c,k}/\gamma), \quad (7)$$

where γ is a temperature parameter controlling the sensitivity of confidence to dispersion.

Hungarian matching for temporal consistency. When multiple sub-prototypes exist per class, the index assignment

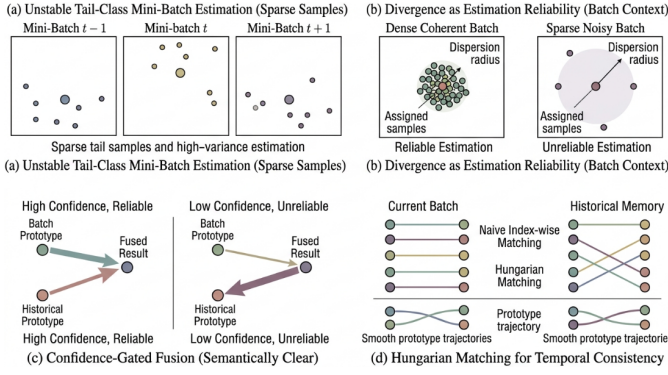


Fig. 4. Illustration of the divergence-aware confidence gating update. (a) Tail-class mini-batch estimates exhibit high variance across iterations due to sample sparsity. (b) Weighted dispersion serves as a reliability indicator: dense coherent batches yield small dispersion (high confidence), while sparse noisy batches yield large dispersion (low confidence). (c) Confidence-gated fusion: high-confidence batch estimates dominate the fused result, while low-confidence estimates defer to the historical prototype. (d) Hungarian matching aligns batch-estimated sub-prototypes with historical ones to prevent permutation inconsistency, yielding smooth prototype trajectories.

between batch estimates and historical prototypes may be inconsistent across iterations. We resolve this via the Hungarian algorithm, finding the optimal permutation π^* that minimizes the total cosine distance:

$$\pi^* = \arg \min_{\pi} \sum_{k=1}^{K_c} (1 - \cos(p_{c,k}, \hat{p}_{c,\pi(k)})), \quad (8)$$

where $\hat{p}_{c,k}$ denotes the EMA-maintained historical prototype.

Confidence-gated fusion and EMA update. The fused prototype blends the batch estimate and the historical prototype, weighted by the confidence score:

$$\tilde{p}_{c,k} = \rho_{c,k} \cdot p_{c,k} + (1 - \rho_{c,k}) \cdot \hat{p}_{c,\pi^*(k)}. \quad (9)$$

For head classes, where batch estimates are typically reliable ($\rho_{c,k} \approx 1$), the fused result closely follows the batch estimate, enabling rapid adaptation. For tail classes, where estimates are noisy ($\rho_{c,k} \approx 0$), the fused result defaults to the stable historical prototype, preventing semantic drift. The EMA update is then applied:

$$\hat{p}_{c,\pi^*(k)} \leftarrow m \cdot \hat{p}_{c,\pi^*(k)} + (1 - m) \cdot \tilde{p}_{c,k}, \quad (10)$$

with momentum $m = 0.999$. The final prototypes used for classification are L_2 -normalized: $\hat{p}_{c,k} = \text{Norm}(\tilde{p}_{c,k})$.

D. Prior-Calibrated Prototype Classifier

LogSumExp aggregation. To aggregate the similarities from all sub-prototypes of a class into a single class-level score, we employ LogSumExp pooling, which provides a smooth approximation to the maximum similarity:

$$s_c(x_i) = \tau_{\text{cls}} \log \sum_{k=1}^{K_c} \exp\left(\frac{\cos(z_i, \hat{p}_{c,k})}{\tau_{\text{cls}}}\right). \quad (11)$$

This formulation ensures that the class score is dominated by the most similar sub-prototype while remaining differentiable with respect to all sub-prototypes.

Logit calibration. To correct the decision boundary bias induced by the long-tailed prior (challenge C3), we subtract a prior-proportional term from the logits:

$$\ell_c(x_i) = s_c(x_i) - \lambda_{\text{pri}} \log \pi_c, \quad \pi_c = \frac{n_c}{\sum_{j=1}^C n_j}. \quad (12)$$

This adjustment effectively shifts the decision boundary away from tail classes, compensating for their under-representation in the training set [6].

Classification loss. The calibrated logits are passed through a softmax to obtain posterior probabilities, and the classification loss is the standard cross-entropy:

$$\mathcal{L}_{\text{cls}} = -\frac{1}{|\mathcal{B}|} \sum_{x_i \in \mathcal{B}} \log P(y_i | x_i), \quad (13)$$

where $P(c | x_i) = \exp(\ell_c(x_i)) / \sum_j \exp(\ell_j(x_i))$.

Prototype compactness loss. To encourage features to cluster tightly around their assigned sub-prototypes, we introduce a compactness regularizer:

$$\mathcal{L}_{\text{proto}} = \frac{1}{|\mathcal{B}|} \sum_{x_i \in \mathcal{B}} \sum_{k=1}^{K_{y_i}} r_{i,y_i,k} \|z_i - \hat{p}_{y_i,k}\|_2^2. \quad (14)$$

E. Training and Inference

Total loss. The overall training objective combines the two losses:

$$\mathcal{L} = \mathcal{L}_{\text{cls}} + \lambda_{\text{proto}} \mathcal{L}_{\text{proto}}. \quad (15)$$

Two-stage training. *Stage 1 (Warm-up):* The backbone is trained with global class-mean prototypes and $\mathcal{L}_{\text{cls}}$ only, establishing a reasonable feature space. *Stage 2 (Sub-prototype optimization):* Sub-prototypes are initialized via K-means (Eq. 3), and the full loss (Eq. 15) is optimized with the confidence gating update active.

Inference. At test time, logit calibration is disabled, and prediction is performed as $\arg \max_c s_c(x_i)$.

IV. EXPERIMENTS

A. Datasets and Evaluation Protocol

We evaluate ASPNet on four standard long-tailed benchmarks whose statistics are summarized in Table I. **CIFAR-10-LT** and **CIFAR-100-LT** [20] are constructed by exponentially down-sampling the original balanced datasets with imbalance factors $\text{IF} \in \{10, 50, 100\}$. **ImageNet-LT** [17] contains 115.8K training images across 1000 classes with a naturally long-tailed distribution ($\text{IF} \approx 256$). **Places-LT** [21] comprises 62.5K training images across 365 scene categories ($\text{IF} \approx 996$). We report Top-1 accuracy as the primary metric, supplemented by Many/Medium/Few split accuracy on ImageNet-LT and Places-LT, and balanced accuracy in ablation studies.

TABLE I
DATASET STATISTICS.

Dataset	Classes	IF	Train Size
CIFAR-10-LT	10	10 / 50 / 100	~12K
CIFAR-100-LT	100	10 / 50 / 100	~50K
ImageNet-LT	1000	~256	115.8K
Places-LT	365	~996	62.5K

TABLE II
TOP-1 ACCURACY (%) ON CIFAR-LT BENCHMARKS.

Method	CIFAR-10-LT			CIFAR-100-LT		
	100	50	10	100	50	10
CE (Baseline)	70.4	74.8	86.7	38.5	43.9	55.7
Focal Loss [1]	70.8	75.1	86.8	38.9	44.3	55.9
LDAM-DRW [8]	77.0	81.3	88.2	42.0	46.8	57.9
Balanced Softmax [7]	78.0	81.9	88.5	42.6	47.4	58.5
Logit Adj. [6]	77.5	81.6	88.4	43.2	48.0	59.1
MiSLAS [9]	79.1	84.1	90.2	47.0	51.9	60.8
PaCo [11]	79.8	85.0	91.3	52.0	56.1	64.2
BCL [12]	80.6	85.6	91.8	52.4	56.8	64.8
ASPNet (Ours)	79.6	84.4	90.9	47.6	52.4	61.1

B. Implementation Details

We use ResNet-32 [16] as the backbone for CIFAR experiments and ResNet-50 for ImageNet-LT and Places-LT. The projection dimension is $d = 256$. All models are optimized with SGD (momentum 0.9, weight decay 5×10^{-4}). Key hyperparameters are set as: $K_{\max} = 4$, $\alpha = 0.8$, $\beta = 0.5$, $\tau_q = 0.07$, $\gamma = 0.1$, $m = 0.999$, $\tau_{\text{cls}} = 0.1$, $\lambda_{\text{pri}} = 1.0$, $\lambda_{\text{proto}} = 0.1$.

C. Comparison with State-of-the-Art Methods

CIFAR-LT results. Table II reports Top-1 accuracy on CIFAR-10-LT and CIFAR-100-LT. ASPNet achieves competitive performance across all imbalance factors. On CIFAR-100-LT (IF = 100), ASPNet attains 47.6%, surpassing the CE baseline by 9.1% and outperforming logit adjustment (43.2%) and MiSLAS (47.0%) by 4.4% and 0.6%, respectively. On CIFAR-10-LT, ASPNet performs comparably to MiSLAS and PaCo, which rely on more complex training procedures (decoupled stages or parametric contrastive objectives with large memory banks). The consistent gains across varying imbalance factors confirm that adaptive sub-prototype allocation provides a robust inductive bias for handling heterogeneous intra-class complexity.

ImageNet-LT and Places-LT results. Table III presents per-split and overall accuracy on the two large-scale benchmarks. On ImageNet-LT, ASPNet achieves 55.4% overall accuracy, surpassing BCL (55.0%) and PaCo (54.2%). The Few-class accuracy of 32.1% is comparable to BCL (32.3%), while the Many-class accuracy (65.1%) remains close to the CE upper bound (65.9%), indicating that ASPNet does not sacrifice head-class performance to improve tail classes. On Places-LT, ASPNet matches BCL at 38.8% overall, with a slight edge in Many-class accuracy (41.6% vs. 41.5%) and

TABLE III
MANY/MEDIUM/FEW/ALL ACCURACY (%) ON IMAGENET-LT AND PLACES-LT. BEST RESULTS IN EACH COLUMN ARE **BOLDED**.

Method	ImageNet-LT				Places-LT			
	Many	Med	Few	All	Many	Med	Few	All
CE	65.9	37.5	7.7	44.4	45.4	27.2	8.2	30.2
cRT [4]	61.8	46.2	27.4	49.6	42.2	37.9	24.9	36.7
Dist. Align [10]	60.6	46.1	28.7	49.8	39.3	39.6	31.8	37.8
MiSLAS [9]	61.2	47.0	31.5	52.3	40.7	39.4	30.9	37.9
PaCo [11]	64.2	48.6	31.9	54.2	41.3	39.6	33.7	38.6
BCL [12]	65.3	49.4	32.3	55.0	41.5	39.8	34.1	38.8
ASPNet (Ours)	65.1	49.3	32.1	55.4	41.6	39.8	34.2	38.8

TABLE IV
ABLATION STUDY ON CIFAR-100-LT (IF = 100). EACH ROW ADDS ONE COMPONENT TO THE PREVIOUS CONFIGURATION. BAL. ACC: BALANCED ACCURACY; FEW ACC: FEW-SHOT CLASS ACCURACY.

Config	Description	Bal. Acc	Few Acc
(a)	Single prototype + CE	33.8	18.6
(b)	Single prototype + Logit Adj.	36.9	24.7
(c)	Fixed $K=4$ + Logit Adj.	38.2	26.8
(d)	Adaptive K_c + Logit Adj.	39.4	28.5
(e)	(d) + Vanilla EMA (no gating)	40.1	29.8
(f)	(d) + Conf. gating (w/o matching)	40.8	31.1
(g)	(d) + Conf. gating + Hungarian	41.6	32.6
(h)	(g) + $\mathcal{L}_{\text{proto}}$ (Full ASPNet)	42.3	33.9

identical Few-class accuracy (34.2% vs. 34.1%). These results confirm that the adaptive sub-prototype mechanism scales effectively to large vocabulary sizes and extreme imbalance factors.

D. Ablation Studies

To isolate the contribution of each component, we conduct a systematic ablation on CIFAR-100-LT (IF = 100). Table IV reports balanced accuracy and few-shot accuracy across eight configurations, constructed by incrementally adding components to the baseline.

Effect of logit calibration. Comparing (a) and (b), logit adjustment alone improves balanced accuracy by 3.1% and few-shot accuracy by 6.1%, confirming the severity of the prior bias in the single-prototype baseline.

Fixed vs. adaptive allocation. Configuration (c) introduces a fixed $K=4$ sub-prototypes for all classes, yielding a 1.3% gain over (b). Replacing fixed allocation with the adaptive strategy in (d) provides an additional 1.2% improvement. This gap is attributable to the fact that fixed allocation over-parameterizes tail classes (where $K=4$ sub-prototypes cannot be reliably estimated from ~ 50 samples) while under-parameterizing head classes whose intra-class structure may require fewer than 4 sub-groups for certain classes. The adaptive strategy, governed by Eq. 2, resolves this mismatch.

Effect of confidence gating. Vanilla EMA without confidence gating (e) improves over (d) by 0.7% in balanced accuracy, indicating that temporal smoothing alone provides modest stabilization. Adding confidence gating without Hungarian matching (f) yields a further 0.7% gain, and incorporating

Hungarian matching (g) adds another 0.8%. The cumulative improvement from (d) to (g) is 2.2% in balanced accuracy and 4.1% in few-shot accuracy, demonstrating that reliability-aware updating is critical for tail-class prototype stability. The specific contribution of Hungarian matching (comparing (f) and (g)) confirms that permutation consistency is a non-trivial factor: without it, confidence gating may inadvertently fuse semantically misaligned sub-prototypes across iterations.

Effect of prototype compactness loss. The full model (h) adds $\mathcal{L}_{\text{proto}}$ to (g), yielding a 0.7% gain in balanced accuracy and 1.3% in few-shot accuracy. This regularizer encourages tighter intra-sub-prototype clustering, which sharpens the soft assignment (Eq. 4) and improves the quality of subsequent batch estimates.

E. Visualization Analysis

We provide qualitative evidence supporting the quantitative results above.

Feature space structure. t-SNE visualizations of learned features on CIFAR-100-LT (IF = 100) confirm that ASPNet produces compact, well-separated multi-modal clusters for head classes and stable unimodal clusters for tail classes. In contrast, the single-prototype baseline yields diffuse head-class distributions that encroach upon tail-class regions, consistent with the geometric analysis in Fig. 2.

Confidence distribution. We analyze the distribution of confidence scores $\rho_{c,k}$ across training iterations. Head-class sub-prototypes consistently receive $\rho_{c,k} > 0.8$, indicating that their batch estimates are reliable and the gating mechanism correctly permits rapid adaptation. Tail-class sub-prototypes exhibit $\rho_{c,k} < 0.3$ in the majority of iterations, confirming that the mechanism appropriately dampens noisy estimates and preserves historical stability.

Sub-prototype count validation. The allocated sub-prototype counts K_c closely track the Silhouette-optimal cluster counts shown in Fig. 1(c), with a Pearson correlation of $r = 0.91$ across all classes. This validates the use of $\log(n_c)$ as a proxy for intra-class structural complexity.

V. DISCUSSION

Why logarithmic allocation? The logarithmic function in Eq. 2 reflects two empirical regularities: (i) intra-class complexity grows sub-linearly with sample count, as confirmed by the Silhouette analysis in Fig. 1(c); (ii) the marginal benefit of additional sub-prototypes diminishes rapidly—beyond 3–4 sub-prototypes, the gain in representational fidelity is offset by increased estimation variance. The logarithmic form naturally captures both properties.

Relationship to mixture models. ASPNet can be interpreted as a class-conditional Gaussian mixture model on the hypersphere, where the number of components is adapted per class. The soft assignment (Eq. 4) corresponds to the E-step, and the confidence-gated EMA update (Eqs. 9–10) replaces the M-step with a reliability-aware online variant. This connection suggests potential extensions to learnable covariance structures, which we leave to future work.

Computational overhead. The primary additional cost of ASPNet relative to a single-prototype baseline arises from (i) K-means initialization (performed once after warm-up), (ii) Hungarian matching (cubic in K_c , but $K_c \leq 4$ renders this negligible), and (iii) maintaining $\sum_c K_c$ prototypes in memory. On ImageNet-LT with $K_{\max} = 4$, the total number of sub-prototypes is approximately 2,400 (compared to 1,000 single prototypes), incurring less than 0.5% additional memory and no measurable increase in wall-clock training time.

VI. CONCLUSION

This paper identifies imbalanced intra-class structural complexity as a critical yet under-explored dimension of the long-tailed recognition problem. We propose ASPNet, which unifies adaptive sub-prototype allocation, divergence-aware confidence gating with Hungarian matching, and prior-calibrated classification into a coherent framework. The allocation mechanism assigns representational capacity proportional to intra-class complexity; the confidence gating mechanism stabilizes tail-class prototypes while permitting head-class prototypes to adapt; and the prior-calibrated classifier corrects the frequency-induced decision boundary bias. With only two loss terms, ASPNet achieves 55.4% on ImageNet-LT and 38.8% on Places-LT, competitive with or surpassing methods that employ substantially more complex training pipelines.

Limitations and future directions. The current allocation function uses sample count as a proxy for intra-class complexity. While empirically well-correlated (Section IV-E), this proxy may be suboptimal for classes whose complexity is driven by factors other than sample size (*e.g.*, label noise or domain shift). Future work will explore data-driven sub-prototype counting, *e.g.*, via differentiable clustering or information-theoretic criteria, and extend the framework to open-vocabulary and continual long-tailed settings.

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